Supplement Material 5. Characteristics of the grey literature studies, main findings, and quality appraisal

Authors, publication year and setting	Participants	Intervention and comparator (prescription formula)	Study design	follow-up duration	Outcomes (tool/test)	Main findings
Leavitt and Flexert (2012) USA	N = 5 Mean age = 56.6 years (38–69) 4FA PTA = 46.7 dB HL (SD 6.7) HL type: SNHL Experienced HA users W EF	Intervention: REM + NAL-R Comparator: Initial fit + manufacturer prescription formula	Within-subject design (Participants were blinded and randomized to the order of the conditions).	In the same session?	**Speech intelligibility in noisy settings** (word-recognition scores in noise [QiuckSIN])	On average, there was a statistical significant advantage for REM fitting over initial fit (about 6.6 dB SNR).
Amlani et al. (2017) USA	Group 1 (experienced HA users) N = 20 (8 M; 12 F) Mean age = 66.8 years (SD 5.2) 4FA PTA = 48 dB HL (2SD 10.6) HL type: SNHL Group 2 (In-the-drawer) N = 20 (13 M; 7 F) Mean age = 66.2 years (SD 4.9) 4FA PTA = 47 dB HL (2SD 8.9) HL type: SNHL Group 3 (first-time) N = 20 (10 M; 10 F) Mean age = 68.5 years (SD 7.3) 4FA PTA = 45 dB HL (2SD 8.5) HL type: SNHL	Intervention: REM + NAL-NL2 Comparator: Initial fit + NAL-NL2	Three paralleled groups (participants were randomized to the treatment groups)	In the same session?	**Speech intelligibility in noisy settings (**connected speech test; speech and noise are spatially separated)	For experienced and in the-drawer groups, there were statistical significant advantages for REM fitting over initial fit. For the first-time group, there was a trend of a better outcome with REM fitting, but was not statistically significant.

Note: N = number of participants; M = male; F = female; SD = standard deviation; 4FA = four frequency average; PTA = pure-tone average; HL = hearing level; SNHL = sensorineural hearing loss; NAL-NL = National acoustic laboratories non-linear; REM = Real-ear measurement; QuickSIN = quick speech in noise test; dB = decibel; SNR = signal-to-noise ratio.

Studies	Downs and Black's quality assessment scores					
	Quality of reporting (10 items), max score = 11	External validity (3 items), max score = 3	Internal validity		Power (1 item), max score = 1	Total (max = 28)
			Study bias (7 items), max score = 7	Confounding and selection bias (6 items), max score = 6		
Leavitt and Flexert (2012)	6	0	6	4	0	16 (fair)
Amlani et al. (2017)	8	1	6	5	0	20 (good)

Studies, and acclimatization periods	Prescription formulae	Condition	Stimulus	Presentation level	Scoring technique
Leavitt and Flexert (2012) No acclimatization	Different prescriptions	Speech in noise	Sentences	At a 57 dB SPL from two loudspeakers	--
Amlani et al. (2017) No acclimatization	Same prescription	Speech in noise	Sentences	At RMS 75 dB SPL Noise presented from three loudspeakers at 66.23 dB SPL each (i.e. at RMS 71 dB SPL when combined together)	Word scoring

Note: RMS = root mean square; dB = decibel; SPL = sound pressure level.